PUSL2021 Computing Group Project

Project Proposal 2024/25

Note-Sharing and Study Collaboration Platform

Group Name: 21

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Introduction

The students in the high-schools and universities are often hampered by getting poor study materials and cannot effectively manage academic commitments. Since the need to negotiate through various different platforms for the purposes of sharing notes, working in collaboration, and communicating in real time, academic life has become fragmented and inefficient. This fragmentation in the learning tools results in students losing their precious opportunities for collaborative learning, which is also crucial for students' success.

By addressing these issues, we would like to propose a system: an integrated Note-Sharing and Study Collaboration Platform that allows note-sharing, formation of study groups, AI-assisted learning, and real-time collaboration in a single convenient platform. This system incentives students for value-added content, with real-time interaction and adaptive learning. This system therefore tends to be more interactive, efficient, and collaborative academically for university students.

Project Objectives

Our project aims to develop a platform that solves specific academic challenges faced by students and undergraduates. The objectives are:

- To identify the problems caused by the fragmentation of tools used for note-sharing, real-time collaboration, and knowledge assessment.
- To design and implement a cohesive platform that provides note-sharing functionality, real-time collaboration tools (such as video conferencing), and AI-driven knowledge support.
- To encourage students to share their notes by providing a system of rewards that includes leaderboards, stars, reviews, and ratings.
- To make individualized learning experiences possible by utilizing performance-tracking tools.
- To enable real-time study sessions within the platform by integrating with the Zoom API.
- To create a user-friendly, accessible, and efficient academic platform that streamlines the student learning process.

Background and Motivation

Nowadays students find it challenging when it comes to managing their academic responsibilities. They often must juggle multiple tools for getting notes, joining study groups or engage in real time discussions. This scattered approach may be confusing and ineffective for students, leading them to miss opportunities and collaborations in learning.

As a result, we came up with our own idea to create a single learning platform which consists of all these features. While tools like Google Classroom, Study Drive, Microsoft Teams, and Course Hero provides solutions partially forcing students to use multiple platforms, our platform addresses this by providing an all-in-one experience. For example, note sharing platforms are not connected with apps like Zoom, therefore they lack the live discussion option.

The inspiration for our project comes from the realization that students need a single platform through which their needs, from note sharing to real time collaborations are met. By creating an interactive rewarding platform with features like real time collaborations, AI powered learning and an incentive-based note sharing system, we address the difficulties in existing platforms and provides a more comprehensive and seamless experience for the users.

Approach/Methodology

Our proposed platform completely provides a student focused experience by combining all essential features into a single app which help them in managing their tasks easily and effectively without jumping between multiple tools.

How our solution stands out in comparison to other platforms:

Most platforms focus on one or two features like note sharing or synchronous collaborations. But we together combine everything- note sharing, live collaboration and customized learning support, saving students time and effort.

The platform's development and deployment will be based on the following methodology:

1. **User centered Design:**
    - **Surveying students:** we plan on getting feedback from students to understand their needs and issues in order to ensure that our platform solves real problems.
    - **Iterative development:** Using Agile development method, we will make our platform in stages, testing and improving while checking on the user feedbacks.
2. **Key Elements:**
    - **Note sharing system:** Users may upload and label their notes based on topics and subjects. After reviewing, others can rate it, so they know which material is the best.

- **Past papers Repository system:** not only focusing on note sharing, we will also include a collection of past papers for both Ordinary Level (OL) and Advanced Level (AL) students for their exam preparations.
- **Reward system:** we plan on implementing a system where contributors earn stars, ratings, and leader-board positions as an incentive for excellent note sharing.
- **AI-powered learning support:** by incorporating an API like ChatGPT, our platform would assist students in receiving answers to their questions, with a personalized learning support.
- **Real time collaboration:** we let our users to host and join live study sessions straight through the platform (the integration of Zoom API will promote real time collaborations). Along with that we will include features like, group document editing and a live chat.

- **Open Contribution Model:** our platform is open for a wide range of contributors, whether it is a current student, a lecturer, a retired teacher, or an educational professional, anyone could upload notes to our site.

3. **Technology Stack:**

   o **Frontend:** The platform will be built using current web technologies as React or Vue.js to provide a responsive and user-friendly design.

   o **Backend:** PostgreSQL or MongoDB will be used to store data in a reliable backend system built with Node.js or Django. Scalability and security will be guaranteed with this setup.

   o **Real-Time Integration:** WebSocket connections are available to manage real-time communication capabilities, which are required for Zoom API and AI integration in order to facilitate instant interactions.

4. **Benefits for Users:**
   - **All in One Access:** students can avoid switching between multiple platforms and manage everything; note sharing, group study, and AI support on a single platform saving up their time.
   - **Motivated Sharing:** We believe social rewards like stars, rating and leader-boards encourage students for active participation in knowledge sharing.
   - **Collaborative Learning:** The capacity to work together, plan study sessions, and get ready for assessments as a group will be improved by real-time collaboration technologies.
   - **Personalized Learning:** Students will be able to learn at their own pace and get support that is specific to their progress thanks to AI-powered help and knowledge assessment tools.

5. **Why This Will Be Effective:**

   Our platform targets actual challenges faced by students daily. Therefore, by creating an **All-in - One platform**, will make sure to address these challenges and make studying easier for students, fostering more collaborative involvement through integrating features they already apply across various platforms into a single system.

Resource Requirements

Hardware:

- **Servers:** A cloud server to host the complete platform. For example AWS, Google cloud or Azure.

Program:

- **Frontend:** HTML, CSS, JavaScript, and React.
- **Backend:** database systems like MySQL or MongoDB, RESTful APIs, and Node.js or Django
- **API Integration:** ChatGPT for AI-powered learning assistance, Zoom API for video conferencing.
  - **Testing Tools:** automated and unit testing tools like Selenium and Jest.
- **Version Control:** Github.

Budgetary Requirements:

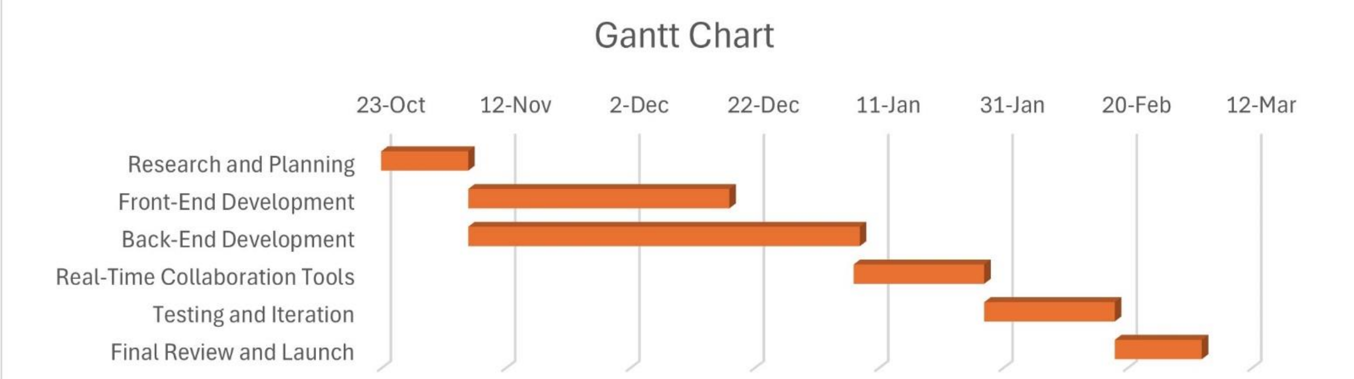
- **Cloud Service Costs:** Monthly payments for bandwidth and hosting usage.
  - **API Access Costs:** Licensing or usage charges for API of ChatGPT and Zoom API.

Future Implements

Looking forward, we are planning to partnership with external businesses and organizations to expand the range of services provided by our study platform. This may include tutoring services, professional development for educators, and additional study materials.

In addition, we plan on having live Zoom lessons for students (one-on-one or group) who wish to engage with the authors of the shared notes or any specific educational professor they wish to clarify their doubts and for further personalized learning experiences.

Project Plan



	Final Review and Launch	Testing and Iteration	Real-Time Collaboration Tools	Back-End Development	Front-End Development	Research and Planning
Start Date	18-Feb	28-Jan	7-Jan	6-Nov	6-Nov	23-Oct
Days to Complete	14	21	21	63	42	14

References

Google Classroom (n.d.). *Google Classroom*. Available at: https://classroom.google.com (Accessed: 23 October 2024).

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